EKRA ISLAM OHI

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EDUCATION

Green University of Bangladesh **Dhaka, Bangladesh**

Bachelor of Science in Computer Science and Engineering | Period: 2022 - 2026 Expected

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, DBMS, OOP

RESEARCH & PUBLICATIONS

Explainable Deep Transfer Learning and Generative Augmentation for Rice Leaf Disease Classification

Ekra Islam Ohi, et al. | Conference Manuscript Submission (Under Peer Review, 2025-2026)

- Evaluated ResNet-18 architectures with DCGAN synthetic patch generation to resolve botanical class imbalance.
- Integrated Grad-CAM saliency mapping to provide transparent, interpretable visual evidence for disease localization.

SELECTED PROJECTS & RESEARCH SYSTEMS

CitationGuard AI - Evidence-Aware Citation Verification System **Python, PyTorch, PubMedBERT**

- Built automated stance verification pipeline evaluating claim grounding against scientific literature under context isolation.

Bangla Document Intelligence - OCR Pipeline & In-Memory Search Engine **Python, EasyOCR, PyMuPDF, Docker**

- Designed high-throughput bilingual PDF OCR engine with instant in-memory text search and containerized deployment.

Butterfly Species Classification & XAI with Grad-CAM **PyTorch, ResNet-18, Streamlit**

- Implemented end-to-end computer vision classifier with native PyTorch hooks for real-time visual explainability heatmaps.

Momin Agro - Smart Dairy & Biogas Management System **React, TypeScript, Supabase, RLS**

- Developed production-ready operational management PWA with PostgreSQL Row Level Security and cattle profiling.

UniPlanner - Academic Course & GPA Forecasting PWA **TypeScript, React, PWA, Service Worker**

- Created offline-first progressive web app with Service Worker caching and Bangladesh UGC/US 4.0 grading support.

TECHNICAL COMPETENCIES

Programming Languages: Python, C, C++, TypeScript, JavaScript, SQL

AI & Machine Learning: PyTorch, Hugging Face Transformers, Scikit-Learn, OpenCV, Grad-CAM, DCGAN, BERT

Tools & Engineering: Docker, Git & GitHub, Linux/Bash, React, Node.js/Fastify, PostgreSQL, Supabase, Vite, PWA

Research Methods: Context Isolation, Stance Detection, OCR Pipelines, Saliency Mapping, Empirical Evaluation

ACADEMIC INTERESTS & FUTURE OBJECTIVES

Focused on Machine Learning Reliability, Natural Language Processing, and Human-Centered AI Systems.

Actively preparing for international graduate studies (M.S. / Ph.D.) and research-driven software engineering roles.